Stan implementation of Sherwood Webb et al 2020

1. Introduction

This is a Stan-based approach for calculating the posterior distributions of ECS from Sherwood, Webb et al. 2020 (hereafter SW20). The original code is slow, taking $\mathcal{O}\,(10 \text{ hrs})$ to calculate a full posterior. The Stan implementation brings it down to $\mathcal{O}\,(10 \text{ secs})$.

By refactoring the posterior calculation in Stan we hope to achieve two goals: 1. increase the speed by two orders of magnitude taking advantage of Stan's Hamiltonian Monte Carlo (HMC) sampler 2. take advantage of Stan's expressive natural language syntax to make the Bayesian framework more user friendly.

Environment:

python environment for running the code can be found in `stan.yml` and can be installed and activated with conda. The most important library is the `cmdstanpy` library which is a python wrapper on stan.

```
conda env create -f stan.yml
conda activate stan
```

Difference from SW20:

There is only one (intentional) difference from SW20's model. Our model does not account for the correlation between historical forcing (F_{hist}) and the forcing associatd with a doubling of CO_2 ($F_{2\times CO_2}$). I tried implementing this correlation and it resulted in very small differences in the posterior of <0.05K at all percentiles (consistent with SW20), but sacrificed code simplicity and ease of reading.

Colab:

A Google colab version of the code can be found here. The colab version may be several commits behind the repository, and is primarily intended as a frictionless demo that does not require installing a new python environment.

This document

The rest of this document outlines the overall statistical model and focuses on a few issues related to the numerical estimation of the posterior. In order to understand the underlying Bayesian framework, readers should familiarize themselves with Sherwood, Webb, et al. 2020, and Marvel and Webb 2025.

2. Stan syntax primer

Stan requires us to define the parameters (i.e. the primary, unconstrained parameters that will be Monte Carlo sampled), any transformed parameters (i.e. parameters deterministically constrained by the primary parameters), and the model formulation. To understand how to formulate a model in Stan, let's take as an example a posterior with two independent parameters $\theta_{1,2}$ and two independent pieces of observational evidence y_1, y_2, which would make the posterior:

$$p(\theta \mid y) = p(y_1, y_2 \mid \theta_1, \theta_2)\, p(\theta_1, \theta_2) = p(y_1 \mid \theta_1, \theta_2)\, p(y_2 \mid \theta_1, \theta_2)\, p(\theta_1)\, p(\theta_2)$$

If, say, the probabilities of $y_1, y_2, \theta_1, \theta_2$ are Gaussian, we could also write this Bayesian model as

$$y_1 \mid \theta_1, \theta_2 \sim \mathcal{N}\left(\mu_{y_1}(\theta_1, \theta_2),\ \sigma_{y_1}(\theta_1, \theta_2)\right)$$

$$y_2 \mid \theta_1, \theta_2 \sim \mathcal{N}\left(\mu_{y_2}(\theta_1, \theta_2),\ \sigma_{y_2}(\theta_1, \theta_2)\right)$$

$$\theta_1 \sim \mathcal{N}\left(\mu_{\theta_1}, \sigma_{\theta_1}\right)$$

$$\theta_2 \sim \mathcal{N}\left(\mu_{\theta_2}, \sigma_{\theta_2}\right).$$

Stan uses two possible formulations. The simplest form of the syntax is based on the second, distributional formulation, where we define the model by simply writing out the four distributions above.

In the back-end, each distributional statement adds to the log-likelihood. For example, since θ_1 has a Gaussian distribution,

$$p_{\theta_1}(\theta_1) = \left(2\pi\sigma_{\theta_1}^2\right)^{-1/2} \exp\left(-\frac{1}{2}\frac{\left(\theta_1 - \mu_{\theta_1}\right)^2}{\sigma_{\theta_1}^2}\right)$$

$$\log p_{\theta_1}(\theta_1) = -\frac{1}{2}\log(2\pi) - \log\sigma_{\theta_1} - \frac{1}{2}\frac{\left(\theta_1 - \mu_{\theta_1}\right)^2}{\sigma_{\theta_1}^2}$$

The distributional statement

$$\theta_1 \sim \mathcal{N}\left(\mu_{\theta_1}, \sigma_{\theta_1}\right)$$

is equivalent to adding $\log p_{\theta_1}(\theta_1)$ to the log-posterior (the `target`, in Stan language),

$$\log\mathcal{L} = \log\mathcal{L} + \log p_{\theta_1}(\theta_1).$$

We can also directly add to log-posterior, and we will need to do that to account for a change of variables if we want to go from uniform S to uniform λ priors without having to change the model structure.

3. The Sherwood-Webb et al. model

3.1 Lines of evidence

SW20 uses four lines of evidence (sec. 7 of SW20): process, historical, cold past climates (LGM) and warm past climates (Pliocene). Under the independence ansatz (eq. 11), the posterior for climate sensitivity, S, is:

$$p(S \mid E_{\text{proc}}, E_{\text{hist}}, E_{\text{paleo}}) \propto p\left(E_{\text{proc}} \mid S\right) p\left(E_{\text{hist}} \mid S\right) p\left(E_{\text{LGM}} \mid S\right) p\left(E_{\text{plio}} \mid S\right) p_S(S)$$

$$p(S \mid E_{\text{proc}}, E_{\text{hist}}, E_{\text{paleo}}) \propto \mathcal{L}_{\text{proc}}(S)\, \mathcal{L}_{\text{hist}}(S)\, \mathcal{L}_{\text{LGM}}(S)\, \mathcal{L}_{\text{plio}}(S)\, p_S(S)$$

The parameters and their coupling equations are as follows:

- **Shared parameters:** S, $F_{2\times}$, ζ, with the feedback parameter defined as $\lambda = -F_{2\times}/S$.
- **Process** (sec. 3): λ_j, 11 feedback parameters, each with a Gaussian distribution.
- **Historical** (sec. 4): $T_{\text{hist}}, F_{\text{hist}}, N_{\text{hist}}, \Delta\lambda$, coupled through equation 6,

$$T_{\text{hist}} = \frac{-\left(F_{\text{hist}} - N_{\text{hist}}\right)}{\lambda - \Delta\lambda}.$$

- **Cold past climates** (sec. 5.2.2): $F_{\text{LGM}}, T_{\text{LGM}}, \alpha, \zeta$, coupled through equation 22,

$$F_{\text{LGM}} = 0.57 \cdot F_{2\times\text{CO}_2} - T_{\text{LGM}} \left(\frac{\lambda}{1+\zeta} + \frac{\alpha}{2} T_{\text{LGM}}\right).$$

- **Warm past climates** (sec. 5.2.3): $F_{\text{plio}}, T_{\text{plio}}, f_{\text{CH}_4}, f_{\text{ESS}}$, coupled through equation 23,

$$T_{\text{plio}} = \frac{-F_{\text{plio}} \cdot \left(1 + f_{\text{CH}_4}\right) \cdot \left(1 + f_{\text{ESS}}\right)}{\lambda/(1+\zeta)}.$$

3.2 Nuisance parameters, primary parameters, and constrained parameters

A Bayesian calculation computes the probability of parameters, θ, given observations y:

$$p(\theta \mid y) \propto p\left(y \mid \theta\right) p(\theta).$$

The SW20 framework is unusual in that it does not provide data points y_j that can be evaluated by plugging them into a parametric likelihood. Rather, it provides distributions of the likelihoods for different lines of evidence, $p\left(E_i \mid S\right)$, that are written in terms of the distributions of certain variables (e.g. $T_{\text{hist}}, F_{\text{hist}}, N_{\text{hist}}, \Delta\lambda$ for the historical evidence). These distributions need to be interpreted as likelihoods rather than frequentist sample distributions (see also Marvel and Webb 2025). Let's take the example of the historical data. For a Bayesian calculation we need to write a likelihood as a probability of the evidence conditional on the parameter S we want to estimate:

$$\mathcal{L}_{\text{hist}}\left(S\right) = p\left(E_{\text{hist}} \mid S\right).$$

However, SW20 does not provide data points, but rather distributions for $(T_{\text{hist}}, N_{\text{hist}}, F_{\text{hist}}, \Delta\lambda)$. The likelihood of the historical evidence can be written as the likelihood of a given combination of variables:

$$\mathcal{L}_{\text{hist}}\left(S\right) = p\left(E_{\text{hist}} \mid S\right) = p(T_{\text{hist}}, N_{\text{hist}}, F_{\text{hist}}, \Delta\lambda).$$

Thus $(T_{\text{hist}}, N_{\text{hist}}, F_{\text{hist}}, \Delta\lambda)$ are not data points but should be viewed as parameters — *nuisance parameters*, since we will integrate them out to get a marginal likelihood or posterior for the parameter of interest S.

Ostensibly the likelihood written above is not a function of S (or λ). However, S links $F, T, N, \Delta\lambda$ such that *for a given value of* S, the four values are not independent. Thus, if we introduce S as a parameter, only three of the other four variables are independent, and one has to become dependent on the others. For example, if we choose temperature as the dependent variable, we can write

$$T(F, N, \Delta\lambda, S) = \frac{-\left(F - N\right)}{\lambda - \Delta\lambda} = \frac{-\left(F - N\right)}{-F_{2\times}/S - \Delta\lambda}.$$

So we can write temperature as a function of λ or of $S, F_{2\times}$ and write the likelihood as:

$$p\left(T, N, F, \Delta\lambda \mid S, F_{2\times}\right) = p(T \mid S, F_{2\times})\, p(N)\, p(F)\, p(\Delta\lambda)$$

$$\mathcal{L} = p\left(T, N, F, \Delta\lambda \mid S, F_{2\times}\right) \propto \exp\left[-\frac{\left(\frac{-(F-N)}{-F_{2\times}/S-\Delta\lambda} - \mu_T\right)^2}{2\sigma_T^2}\right] \exp\left[-\frac{\left(N - \mu_N\right)^2}{2\sigma_N^2}\right] \exp\left[-\frac{\left(\Delta\lambda - \mu_{\Delta\lambda}\right)^2}{2\sigma_{\Delta\lambda}^2}\right] \exp$$

Thus temperature is now a *constrained* parameter that is a deterministic function of other parameters. In practice we will sample the primary (unconstrained) parameters $(N_{\text{hist}}, F_{\text{hist}}, \Delta\lambda, F_{2\times}, S)$ via a Monte Carlo technique, and we'll compute the constrained parameter T_{hist} (transformed parameter in Stan's language) from the primary parameters, before plugging it into the likelihood distribution for T_{hist}. Note that we do not necessarily need to make T_{hist} the constrained parameter, and we could make N_{hist}, F_{hist}, or even $\Delta\lambda$ instead.

3.3 The full model:

$$p(S \mid E) \propto \mathcal{L}_{\text{proc}}(S)\,\mathcal{L}_{\text{hist}}(S)\,\mathcal{L}_{\text{LGM}}(S)\,\mathcal{L}_{\text{plio}}(S)\;p(\zeta)\,p(F_{2\times})\,p_S(S)$$

$$\begin{aligned}
\mathcal{L}_{\text{proc}} &= p(\lambda \mid S, F_{2\times}) \\
\mathcal{L}_{\text{hist}} &= p\left(T_{\text{hist}} \mid S, N_{\text{hist}}, \Delta\lambda, F_{\text{hist}}\right)\; p\left(N_{\text{hist}}\right)\; p\left(F_{\text{hist}}\right)\; p(\Delta\lambda) \\
\mathcal{L}_{\text{LGM}} &= p\left(F_{\text{LGM}} \mid S, T_{\text{LGM}}, \zeta, \alpha\right)\; p\left(T_{\text{LGM}}\right)\; p(\alpha) \\
\mathcal{L}_{\text{plio}} &= p\left(T_{\text{plio}} \mid S, F_{\text{plio}}, f_{\text{ESS}}, f_{\text{CH}_4}\right)\; p(F_{\text{plio}})\, p\left(f_{\text{ESS}}\right)\; p(f_{\text{CH}_4})
\end{aligned}$$

4. Implementation Issues

4.1 Choices of constrained parameters

The coupling equations link S with the nuisance parameters used in the likelihood formulation. However, it is up to us to choose which of the parameters becomes the constrained parameter. For example, the following two formulations for the historical likelihood are equivalent:

$$\mathcal{L}_{\text{hist}}(S) = p(T \mid S, F_{2\times})\, p(N)\, p(F)\, p(\Delta\lambda); \quad \text{with} \quad T = \frac{-(F-N)}{\lambda - \Delta\lambda} = \frac{-(F-N)}{-F_{2\times}/S - \Delta\lambda}$$

$$\mathcal{L}_{\text{hist}}(S) = p(N \mid S, F_{2\times})\, p(T)\, p(F)\, p(\Delta\lambda); \quad \text{with} \quad N = F - T\left(-F_{2\times}/S - \Delta\lambda\right).$$

However, there are a couple of considerations that can help us decide which parameter to choose as the constrained parameter:

1. Pick the formulation that doesn't require solving a non-trivial inverse. The LGM equation (eq. 22) is quadratic in T_{LGM} because of the state-dependence term $\frac{\alpha}{2}(\Delta T)^2$:

$$\frac{\alpha}{2}T_{\text{LGM}}^2 + \frac{\lambda}{1+\zeta}T_{\text{LGM}} - \left[0.57\,F_{2\times} - F_{\text{LGM}}\right] = 0.$$

Treating T_{LGM} as the S-coupling would require solving this for T_{LGM} via the quadratic formula, which has (i) two roots and a branch-picking problem, (ii) a $1/\alpha$ singularity that is hit constantly because $\alpha \sim \mathcal{N}(0.1, 0.1)$ passes through zero, and (iii) a discriminant that can go negative near the boundary of physical configurations, producing NaN in autodiff. Going the other way — sampling T_{LGM} as primary and computing F_{LGM} forward — is a single evaluation of a smooth polynomial, with no branches and no singularities. So F_{LGM} is forced to be the S-coupling for LGM.

2. Don't put a kinked / discontinuous likelihood on a transformed parameter. F_{hist} has Sherwood's asymmetric Gaussian observation (different σ above and below μ_F); the log-density jumps by $\log(\sigma_{\text{lo}}/\sigma_{\text{hi}}) \approx 0.747$ at $F_{\text{hist}} = \mu_F$. In the Stan Hamiltonian Monte Carlo (HMC) sampler, if F_{hist} were the historical S-coupling (transformed parameter receiving the asymmetric likelihood), that discontinuity would propagate through the Hamiltonian energy-balance equation into the joint log-density in primary-parameter space, biasing HMC. So we do not want to use F_{hist}. We could choose either N_{hist} or $\Delta\lambda$ instead.

The primary unconstrained parameters define Stan's sampling space, and they are declared in the `parameters` block. We will use the following unconstrained parameters:

$$\theta = [S,\, F_{2\times},\, \zeta,\, F_{\text{hist}},\, N_{\text{hist}},\, \Delta\lambda,\, T_{\text{LGM}},\, \alpha,\, \text{CO}_{2,\text{plio}},\, f_{\text{CH}_4},\, f_{\text{ESS}}].$$

Stan allows us to define any number of constrained parameters, which are declared in the `transformed parameters` block. These will consist not only of the three parameters defined by the three coupling equations needed to write the likelihood, but also any other constrained parameter that might make the model formulation easier, or that we want to output. For example, we may want to define $\lambda = -F_{2\times}/S$ or $F_{\text{plio}} = \log_2\left(\text{CO}_{2,\text{plio}}/284\right) \cdot F_{2\times}$. In this model formulation, I've chosen T_{hist}, F_{LGM}, and T_{plio}. Any other formulation that does not violate rules 1 and 2 should work.

4.2 Avoiding biases from the historical aerosol forcing formulation

tl;dr: F_{hist} has an asymmetric Gaussian likelihood. Sampling F_{hist} directly through Stan's creates a finite log-density jump at $F = \mu_F$ that biases HMC. We sample an unconstrained z-score F_z ~ N(0,1) and define F_hist = mu_F + F_z * sigma(sign(F_z)) as a transformed parameter; this induces exactly the same asymmetric distribution but is smooth in sampling space.

SW20- encode the 5/50/95 percentiles of F_{hist} via two half-Gaussians glued at μ_F,

$$p_F(F) \;=\; \begin{cases} \dfrac{1}{\sigma_{\text{lo}}\sqrt{2\pi}}\exp\Big(-\tfrac{1}{2}\big(\tfrac{F-\mu_F}{\sigma_{\text{lo}}}\big)^2\Big), & F < \mu_F, \\ \dfrac{1}{\sigma_{\text{hi}}\sqrt{2\pi}}\exp\Big(-\tfrac{1}{2}\big(\tfrac{F-\mu_F}{\sigma_{\text{hi}}}\big)^2\Big), & F \geq \mu_F, \end{cases} \qquad \sigma_{\text{lo}} = 1.131,\ \sigma_{\text{hi}} = 0.535.$$

Each half integrates to $\frac{1}{2}$, so p_F is a proper PDF, but it is discontinuous at $F = \mu_F$, where the log-density jumps by

$$\Delta \log p_F \;=\; \log(\sigma_{\text{lo}}/\sigma_{\text{hi}}) \;\approx\; 0.747.$$

HMC's leapfrog integrator assumes a continuously differentiable target, and the jump in (log-)density leads to a slight bias in the posterior.

Fix. Introduce $z \sim \mathcal{N}(0,1)$ and define F deterministically:

$$F(z) \;=\; \mu_F \;+\; \begin{cases} z\,\sigma_{\text{lo}}, & z < 0, \\ z\,\sigma_{\text{hi}}, & z \geq 0. \end{cases}$$

Under this map, F has exactly the asymmetric Gaussian density of Section 4.2: for $F < \mu_F$, we have $z = (F - \mu_F)/\sigma_{\text{lo}}$ and $|\mathrm{d}F/\mathrm{d}z| = \sigma_{\text{lo}}$, so

$$p(F) \;=\; \frac{p_z(z)}{|\mathrm{d}F/\mathrm{d}z|} \;=\; \frac{1}{\sigma_{\text{lo}}\sqrt{2\pi}}\exp\Bigl(-\tfrac{1}{2}\bigl(\tfrac{F-\mu_F}{\sigma_{\text{lo}}}\bigr)^2\Bigr)$$

and analogously on the other side. Critically, in the *sampling space* (z-space) the prior is $\mathcal{N}(0,1)$, which is everywhere smooth. The map $F(z)$ is continuous (both branches give $F = \mu_F$ at $z = 0$); only its derivative has a kink at $z = 0$, which HMC handles correctly. The discontinuous jump in log-density that confused leapfrog in F-space has been moved to a kink in the *transformation*, where it is benign.

4.3 Switching between uniform-λ and uniform-S priors

SW20 presents two choices of priors: uniform in λ and uniform in S. If we write the model in S-space, with S as a primary parameter, declaring `S` with `<lower=A, upper=B>` is equivalent to a prior on S that is uniform on (A, B).

Setting $\lambda = -F_{2\times}/S$ at fixed $F_{2\times}$, the change of variable gives

$$\left.\frac{\partial S}{\partial \lambda}\right|_{F_{2\times}} = \frac{F_{2\times}}{\lambda^2} = \frac{S^2}{F_{2\times}}, \qquad \left.\frac{\partial \lambda}{\partial S}\right|_{F_{2\times}} = \frac{F_{2\times}}{S^2}.$$

The implicit prior on $(\lambda, F_{2\times})$ is then

$$p(\lambda, F_{2\times}) = p_S\left(S(\lambda, F_{2\times})\right) \left.\frac{\partial S}{\partial \lambda}\right|_{F_{2\times}} p_{F_{2\times}}(F_{2\times}) \;=\; p_S\left(S(\lambda, F_{2\times})\right)\cdot \frac{F_{2\times}}{\lambda^2}\cdot p_{F_{2\times}}(F_{2\times}) \;\propto\; \frac{F_{2\times}}{\lambda^2}\, p_{F_{2\times}}(F_{2\times}),$$

i.e. *non-uniform* in λ at fixed $F_{2\times}$. This is the "U_S" prior of Sherwood (sec. 7.2): uniform on S.

To recover the paper's U_λ prior (uniform on each component feedback λ_i, which by convolution is approximately uniform on λ) we can either rewrite the model to use λ as a primary parameter and give it a uniform prior, or make the change of variables by adding

```
target += log(F_2xCO2) - 2 * log(S);
```

This multiplies the joint by $S^2/F_{2\times}$, giving an implicit prior on $(\lambda, F_{2\times})$ proportional to $p_{F_{2\times}}(F_{2\times})$ — i.e. uniform in λ at every $F_{2\times}$ slice, with $F_{2\times}$ retaining its Gaussian prior. This matches Sherwood's U_λ exactly (uniform-λ $\times$ Gaussian-$F_{2\times}$).